PRECOMPOSITION ON CONTINUOUS ALTERNATING MAPS IS NOT ANALYTIC IN CHARACTERISTIC 2

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Abstract. For normed spaces E, E', F over a nontrivially normed field k and a finite ι, precomposition $f \mapsto (m \mapsto m \circ (f, \dots, f))$ sends continuous linear maps $E \to E'$ to continuous linear maps between spaces of continuous alternating maps. We show that it is *not* analytic in general. If k is a complete non-Archimedean field of characteristic 2 that is not spherically complete, then for $E = E' = \ell^\infty$ and $|\iota| = 2$ there is a Banach space F over k for which this map is not analytic at 0. The proof is a finite parity identity for determinants in characteristic 2, glued along ℓ^∞. The result is formalized in Lean 4/Mathlib.

1. Introduction

Let k be a nontrivially normed field. For normed spaces E, E', F over k and a finite ι we write $\mathcal{L}(E, E')$ for the space of continuous linear maps with the operator norm, and $\mathrm{Alt}^\iota(E, F)$ for the normed space of continuous alternating ι-linear maps $E^\iota \to F$; we write Alt^2 when $|\iota| = 2$.

Analytic functors. A map $f\colon X \to Z$ between normed spaces over k is **analytic at** $x_0 \in X$ if there are bounded n-linear maps $p_n\colon X^n \to Z$ and a radius $r > 0$ with $\sup_n \|p_n\| r^n < \infty$ such that $f(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h)$ whenever $\|h\| < r$. It is **analytic**, or C^ω, if it is analytic at every point of X; this is the definition used in Mathlib [5].

Lang [4, Ch. III, §4] calls a functor T on Banach spaces **of class C^p** if $f \mapsto T(f)$ is C^p. Similarly, we call a contravariant endofunctor T on the category of normed spaces over k **analytic** if for all E, E' the map

$$\mathcal{L}(E, E') \to \mathcal{L}\big(T(E'), T(E)\big), \qquad f \mapsto T(f),$$

is analytic. For a normed space F and a finite ι, the assignment $E \mapsto \mathrm{Alt}^\iota(E, F)$, $f \mapsto (m \mapsto m \circ (f, \dots, f))$, is a contravariant functor. The assignment is also functorial in F: a continuous linear map $g\colon F \to F'$ acts by $m \mapsto g \circ m$. This action is linear in g, hence analytic, whereas the action of f is not linear in f. So the bifunctor Alt^ι is analytic if and only if the functor $\mathrm{Alt}^\iota(-, F)$ is analytic for every F.

Motivation: alternating bundles in Mathlib. In Mathlib, manifolds and vector bundles are modelled on normed spaces (of possibly infinite dimension) over an arbitrary nontrivially normed field k, and their regularity is an element $n \in \mathbb{N} \cup \{\infty, \omega\}$, where C^ω means analytic. Differential forms require the bundle of continuous alternating maps between two C^n vector bundles to be a C^n vector bundle again (an instance of `ContMDiffVectorBundle`). Its transition functions are those of the given bundles with Alt^ι applied, so this comes down to the regularity of the functor $\mathrm{Alt}^\iota(-, F)$, i.e. of the precomposition map

`ContinuousAlternatingMap.compContinuousLinearMapCLM`.

Two cases have been settled, in pull requests to Mathlib.

- If $|\iota|!$ is invertible in k then $\mathrm{Alt}^\iota(-,F)$ is analytic, by polarisation. In Mathlib this is done in characteristic 0 by Gouëzel [1], who uses it to construct the bundle for every n in characteristic 0 [2]; he remarks that a variant of the argument, not included there, covers finite-dimensional spaces in positive characteristic, but not the general case.
- Over every k, with no hypothesis on the characteristic or the dimension, $\mathrm{Alt}^\iota(-,F)$ is of class C^n for every $n \in \mathbb{N} \cup \{\infty\}$ [6].

This leaves open the case where $|\iota|!$ is not invertible in k and the spaces are infinite-dimensional: is $\mathrm{Alt}^\iota(-,F)$ analytic there? The following theorem provides a counterexample.

Theorem 1. *Let k be a complete non-Archimedean field of characteristic 2 which is not spherically complete. There is a Banach space F over k such that the functor $\mathrm{Alt}^2(-,F)$ is not analytic.*

2. Statement

Notation. For a polynomial f and a monomial μ, $\operatorname{coeff}_\mu f$ is the coefficient of μ in f. We write $c_0 \subset \ell^\infty$ for the null sequences, $\mathbf{1}_S$ for the indicator of a set S, and $\mathbf{1} = \mathbf{1}_{\mathbb{N}}$.

Throughout, k is a field which is complete for a nontrivial absolute value $|\cdot|$ satisfying the **ultrametric inequality** $|x+y| \le \max(|x|,|y|)$; we call such a field **non-Archimedean**. The sup norms used below inherit the inequality. We assume that

(K1) $\operatorname{char} k = 2$;
(K2) k is not spherically complete: there is a sequence of closed balls $B(c_n, r_n)$ such that $|c_{n+1} - c_n| \le r_n$, $r_0 > r_1 > \cdots$ and $\bigcap_n B(c_n, r_n) = \emptyset$.

The balls in (K2) are balls in k, and they are nested by the ultrametric inequality. Such fields exist; an explicit one is given in Appendix A. Let $E = \ell^\infty = \ell^\infty(\mathbb{N}, k)$ with the sup norm, a Banach algebra under pointwise multiplication; write $m_a \in \mathcal{L}(E,E)$ for multiplication by $a \in E$, so that $\|m_a\| = \|a\|$, and e_i for the unit vectors. For $x, y \in E$ let

$$W(x,y) := x \otimes y + y \otimes x \in \ell^\infty(\mathbb{N} \times \mathbb{N}, k), \qquad W(x,y)_{pq} = x_p y_q + x_q y_p,$$

and let $F \subseteq \ell^\infty(\mathbb{N} \times \mathbb{N}, k)$ be the closed linear span of all $W(x,y)$, a Banach space. Let

$$A \colon \mathcal{L}(E,E) \to \mathcal{L}\big(\mathrm{Alt}^2(E,F), \mathrm{Alt}^2(E,F)\big), \qquad A(f)(m) = m \circ (f,f).$$

Theorem 2. *A is not analytic at 0. In particular A is not C^ω.*

The map A is the action on $\mathcal{L}(E,E)$ of the functor $\mathrm{Alt}^2(-,F)$, so Theorem 2 implies Theorem 1. Sections 3–6 prove Theorem 2 in four steps.

3. Step 1: analyticity forces a bilinear lift

Recall that $A \colon \mathcal{L}(E,E) \to \mathcal{L}(\mathrm{Alt}^2(E,F), \mathrm{Alt}^2(E,F))$ is given by $A(f)(m) = m \circ (f,f)$, so that $A(tf) = t^2 A(f)$. Steps 1 and 2 do not use (K2): until Step 3, k may be any non-Archimedean field of characteristic 2.

Lemma 3. *Let $A \colon X \to Z$ be a map between normed spaces with $A(tf) = t^2 A(f)$. If A is analytic at 0 then there is a bounded bilinear $P \colon X \times X \to Z$ with $P(f,f) = A(f)$ for all f.*

Proof. Let $\sum_n p_n$ be a power series of A at 0, and fix $f \in X$. For scalars $t \in k$ with $|t|$ small,

$$t^2 A(f) = A(tf) = \sum_n t^n \, p_n(f, \ldots, f).$$

Both sides are power series in the variable $t \in k$ with coefficients in Z. The coefficients of a convergent power series in one variable are unique over any nontrivially normed field, so comparing the coefficients of t^2 gives $p_2(f,f) = A(f)$. Take $P = p_2$. □

So assume that a bounded bilinear P with $P(f,f) = A(f)$ exists. The map $W\colon E \times E \to F$ is bilinear, it is alternating because $W(x,x) = 2\,x \otimes x = 0$, and it is continuous because $\|W(x,y)\| \le \|x\|\|y\|$ by the ultrametric inequality; so $W \in \mathrm{Alt}^2(E,F)$ with $\|W\| \le 1$. Put, for $a, b, x, y \in E$,

$$\Phi(a,b;x,y) := P(m_a, m_b)(W)(x,y) \in F.$$

Then Φ is 4-linear, and since $\|m_a\| = \|a\|$ and $\|W\| \le 1$ we have

$$\|\Phi(a,b;x,y)\| \le C\|a\|\,\|b\|\,\|x\|\,\|y\|, \qquad C := \|P\|.$$

Here $C \ge 1$, because $P(\mathrm{id},\mathrm{id}) = A(\mathrm{id})$ is the identity map of $\mathrm{Alt}^2(E,F) \neq 0$. Moreover

$$\Phi(a,b;x,x) = 0, \qquad \Phi(a,a;x,y) = W(ax,ay). \tag{1}$$

Since $2 = 0$, the first identity gives the symmetry $\Phi(a,b;x,y) = \Phi(a,b;y,x)$. We will contradict the existence of such a Φ.

4. Step 2: a finite parity identity

Fix a finite set $I \subset \mathbb{N}$ of even size $2m$ and $a', b' \in E$ with $\|a'\|, \|b'\| \le 1$. Let X_i, Y_i $(i \in I)$ be indeterminates. Expanding Φ multilinearly in $x = \sum_{i\in I} X_i e_i$ and $y = \sum_{i\in I} Y_i e_i$ defines, for any fixed $p, q \in I$, a polynomial

$$G_{pq} := \Phi(a' + x,\, b' + y;\, a' + x,\, b' + y)_{pq} \in k[X_i, Y_i]_{i\in I}$$

in the polynomial ring in these $4m$ variables, which we abbreviate to $k[X,Y]$. The values of the G_{pq} at points $x, y \in k^I$ (vectors supported on I) are the entries of an $I \times I$ matrix $G(x,y)$. We also put

$$g_{pq} := \Phi(x,y;x,y)_{pq} = \sum_{i,j,k,l\in I} \Phi(e_i,e_j;e_k,e_l)_{pq}\, X_iY_jX_kY_l \in k[X,Y].$$

Proposition 4. *The matrix $G = (G_{pq})_{p,q\in I}$ is symmetric with zero diagonal: $G_{pq} = G_{qp}$ and $G_{pp} = 0$ in $k[X,Y]$.*

Proof. Every element of F is a symmetric matrix with zero diagonal, since $W(x,y)_{pq} = W(x,y)_{qp}$ and $W(x,y)_{pp} = 2x_py_p = 0$, and both properties pass to the closed span. The coefficients of G_{pq} are the (p,q) entries of values of Φ, which lie in F. □

Proposition 5. *The polynomial g_{pq} is homogeneous of degree 2 in the X_i and of degree 2 in the Y_i, and $G_{pq} - g_{pq}$ has total degree ≤ 3. In particular G_{pq} has total degree ≤ 4.*

Proof. By 4-linearity, G_{pq} is a sum of 16 terms, according to whether each of the four slots is filled by the constant vector (a' or b') or by the variable one (x or y), and the total degree of a term is the number of its variable slots. So g_{pq}, which has two slots with x and two with y, is exactly the degree 4 part of G_{pq}, and $G_{pq} - g_{pq}$ has total degree ≤ 3. □

Proposition 6. *For $p \neq q$ we have, in $k[X,Y]$,*

$$g_{pq}(Y,X) = g_{pq}(X,Y) + X_p^2Y_q^2 + X_q^2Y_p^2 \qquad \text{and} \qquad g_{pq}(X,X) = 0.$$

Proof. The field k is infinite, since it has a nontrivial absolute value, and a polynomial over an infinite field which vanishes at every point is zero. So it suffices to check these identities at

points $x, y \in k^I$. By the symmetry in the last two slots, $g(y,x) = \Phi(y,x;y,x) = \Phi(y,x;x,y)$, so by bilinearity in the first two slots and (1)

$$\begin{aligned} g(x,y) + g(y,x) &= \Phi(x+y, x+y; x, y) + \Phi(x,x;x,y) + \Phi(y,y;x,y) \\ &= W\big((x+y)x, (x+y)y\big) + W(x^2, xy) + W(xy, y^2) \\ &= \big[W(x^2,xy) + W(x^2,y^2) + W(xy,xy) + W(xy,y^2)\big] \\ &\quad + W(x^2,xy) + W(xy,y^2) \\ &= W(x^2, y^2), \end{aligned}$$

where the last step uses $2 = 0$ and $W(xy, xy) = 0$. This is the matrix whose (p,q) entry is $x_p^2 y_q^2 + x_q^2 y_p^2$. Finally $g(x,x) = W(x^2, x^2) = 0$. □

Let τ be the automorphism of $k[X,Y]$ swapping $X_i \leftrightarrow Y_i$, and δ the substitution $Y_i := X_i$.

Lemma 7 (balanced coefficients)**.** *Let $f \in k[X,Y]$ with $\tau f = f$. For every $S \subseteq I$ the coefficient of $\prod_{i\in S} X_i Y_i$ in f equals the coefficient of $\prod_{i\in S} X_i^2$ in δf.*

Proof. The coefficient of $\prod_S X_i^2$ in δf is the sum of the coefficients in f of all monomials $X^\alpha Y^\beta$ with $\alpha + \beta = 2 \cdot \mathbf{1}_S$. The involution $(\alpha, \beta) \mapsto (\beta, \alpha)$ of this set of monomials preserves coefficients (as $\tau f = f$), and its only fixed point is $\alpha = \beta = \mathbf{1}_S$; all other terms cancel in pairs. Thus the coefficient of $\prod_S X_i^2$ in δf is equal to the coefficient of $X^{\mathbf{1}_S} Y^{\mathbf{1}_S} = \prod_{i\in S} X_i Y_i$ in f. □

A **perfect matching** of I is a partition of I into m unordered pairs $\{p,q\}$, its edges; for example $\{1,2,3,4\}$ has the three perfect matchings $12|34$, $13|24$ and $14|23$. Since $G_{pq} = G_{qp}$ (Proposition 4), the product of the G_{pq} over the edges of a perfect matching is well defined.

Proposition 8. *Let M be a perfect matching of I and $\nu := \prod_{i\in I} X_i Y_i$. Then*

$$\operatorname{coeff}_\nu \prod_{\{p,q\}\in M} G_{pq} = 1.$$

Proof. Write $G_{pq} = g_{pq} + c_{pq}$, where g_{pq} is homogeneous of total degree 4 and c_{pq} has total degree ≤ 3 (Proposition 5). Expanding the product over the m edges of M gives $\prod_M g_{pq}$ plus terms that contain at least one factor c_{pq}, and each such term has total degree $\le 3 + 4(m-1) = 4m-1$. The monomial ν has total degree $4m$, so only $\prod_M g_{pq}$ contributes to its coefficient:

$$\operatorname{coeff}_\nu \prod_M G_{pq} = \operatorname{coeff}_\nu \prod_M g_{pq}.$$

Put $g'_{pq} := g_{pq} + X_p^2 Y_q^2$. By Proposition 6,

$$\tau g'_{pq} = g'_{pq} \qquad \text{and} \qquad \delta g'_{pq} = X_p^2 X_q^2.$$

A product containing a factor $X_p^2 Y_q^2$ has no squarefree monomial, so replacing every g_{pq} by g'_{pq} does not change the coefficient of ν. By Lemma 7, applied to the τ-invariant product $\prod_M g'_{pq}$, this coefficient is the coefficient of $\prod_{i\in I} X_i^2$ in

$$\delta \prod_M g'_{pq} = \prod_{\{p,q\}\in M} X_p^2 X_q^2 = \prod_{i\in I} X_i^2,$$

which is exactly 1. □

Lemma 9. *Let A be a symmetric matrix with zero diagonal over a commutative ring of characteristic 2, of even size $2m$. Then $\det A = \sum_M \prod_{\{p,q\}\in M} A_{pq}^2$, the sum being over perfect matchings; and the number of perfect matchings is $(2m-1)!!$.*

Proof. In characteristic 2 we have $\det A = \sum_\sigma \prod_i A_{i\sigma(i)}$. As A is symmetric, σ and σ^{-1} give the same product, so the permutations with $\sigma \neq \sigma^{-1}$ cancel in pairs and only the involutions remain. An involution with a fixed point i contributes the factor $A_{ii} = 0$. An involution without fixed points is a product of m disjoint transpositions $(p\ q)$; the pairs $\{p,q\}$ form a perfect matching M, every perfect matching arises exactly once in this way, and therefore

$$\det A = \sum_M \prod_{\{p,q\}\in M} A_{pq}A_{qp} = \sum_M \prod_{\{p,q\}\in M} A_{pq}^2.$$

Finally, pairing the smallest index with one of the $2m-1$ others and continuing with the remaining indices shows that there are $(2m-1)(2m-3)\cdots 3\cdot 1 = (2m-1)!!$ perfect matchings. □

Lemma 10 (hypercube sum)**.** *Let R be a commutative ring of characteristic 2 and let $f \in R[Z_1,\dots,Z_N]$ have total degree $\le N$. Then $\sum_{z\in\{0,1\}^N} f(z)$ is the coefficient of $Z_1Z_2\cdots Z_N$ in f.*

Proof. By linearity we may take f to be a monomial Z^α. At a point $z \in \{0,1\}^N$ it takes the value 1 if $z_i = 1$ for all i in the support of α, and 0 otherwise; so the sum is $2^{N-|\operatorname{supp}\alpha|}$, which vanishes in R unless every variable occurs in Z^α. The only monomial of degree $\le N$ in which all N variables occur is $Z_1\cdots Z_N$. □

Corollary 11. *We have*

$$\sum_{x,y\in\{0,1\}^I} \det G(x,y) = 1,$$

and consequently there are $x, y \in \{0,1\}^I$ with $|\det G(x,y)| \ge 1$.

Proof. Each G_{pq} has total degree ≤ 4 (Proposition 5), so for a perfect matching M the product $\prod_{\{p,q\}\in M} G_{pq}$ has total degree $\le 4m$. By Lemma 10 with $N = 4m$, the number of variables of $k[X,Y]$, and by Proposition 8,

$$\sum_{x,y\in\{0,1\}^I} \prod_{\{p,q\}\in M} G_{pq}(x,y) = \operatorname{coeff}_\nu \prod_{\{p,q\}\in M} G_{pq} = 1, \qquad \text{where } \nu = \prod_{i\in I} X_iY_i.$$

For each pair x, y the matrix $G(x,y)$ is symmetric with zero diagonal (Proposition 4), so Lemma 9 applies to it. In characteristic 2 we have the Frobenius endomorphism $a \mapsto a^2$ of k. Hence

$$\begin{aligned}\sum_{x,y\in\{0,1\}^I} \det G(x,y) &= \sum_{x,y\in\{0,1\}^I} \sum_M \prod_{\{p,q\}\in M} G_{pq}(x,y)^2 \\ &= \sum_M \Big(\sum_{x,y\in\{0,1\}^I} \prod_{\{p,q\}\in M} G_{pq}(x,y)\Big)^2 = \sum_M 1^2 = (2m-1)!! = 1,\end{aligned}$$

because $(2m-1)!!$ is odd. By the ultrametric inequality, $1 \le \max_{x,y\in\{0,1\}^I} |\det G(x,y)|$. □

5. Step 3: locality on ℓ^∞

This step uses (K2) and the completeness of k, which are standing hypotheses of §2, but not the characteristic. Recall (K2): there are centres $c_n \in k$ and radii $r_0 > r_1 > \cdots > 0$ with $|c_{n+1}-c_n| \le r_n$ such that no element of k lies in all the balls $B(c_n, r_n) = \{a \in k : |a-c_n| \le r_n\}$. Recall also that $e_i \in \ell^\infty$ are the unit vectors. For $x \in \ell^\infty$ write $x|_{\ge N}$ for x with its coordinates below N set to 0.

Proposition 12. *A bounded linear functional φ on ℓ^∞ with $\varphi(e_i) = 0$ for all i is zero.*

Proof. Suppose $\varphi \neq 0$. Since $r_{n+1} < r_n$ we have $(r_{n+1}/r_n)\,\|\varphi\| < \|\varphi\|$, so by the definition of the operator norm there is $u_n \in \ell^\infty$ with $|\varphi(u_n)| > (r_{n+1}/r_n)\,\|\varphi\|\,\|u_n\|$. Put $s_n := (c_{n+2} - c_{n+1})\,u_n/\varphi(u_n)$. Then

$$\varphi(s_n) = c_{n+2} - c_{n+1} \qquad \text{and} \qquad \|s_n\| = |c_{n+2} - c_{n+1}|\,\frac{\|u_n\|}{|\varphi(u_n)|} \le r_{n+1} \cdot \frac{r_n}{r_{n+1}\|\varphi\|} = \frac{r_n}{\|\varphi\|}.$$

The difference $s_n - s_n|_{\ge n}$ is finitely supported, so by linearity $\varphi(s_n - s_n|_{\ge n}) = 0$; hence $\varphi(s_n|_{\ge n}) = c_{n+2} - c_{n+1}$. Moreover $\|s_n|_{\ge n}\| \le \|s_n\| \le r_n/\|\varphi\|$. Define $y := \sum_k s_k|_{\ge k}$ coordinatewise; the sum is finite in every coordinate i because $s_k|_{\ge k}$ vanishes at i when $k > i$. By the ultrametric inequality $\|y\| \le \max_k \|s_k\| \le r_0/\|\varphi\|$, so $y \in \ell^\infty$. Likewise $\|y - \sum_{k<n} s_k|_{\ge k}\| = \|\sum_{k\ge n} s_k|_{\ge k}\| \le \max_{k\ge n}\|s_k\| \le r_n/\|\varphi\|$. Hence, using the telescoping sum $\sum_{k<n} \varphi(s_k|_{\ge k}) = c_{n+1} - c_1$,

$$|\varphi(y) + c_1 - c_{n+1}| = \Big|\varphi\Big(y - \sum_{k<n} s_k|_{\ge k}\Big)\Big| \le \|\varphi\|\,\Big\|y - \sum_{k<n} s_k|_{\ge k}\Big\| \le \|\varphi\| \cdot \frac{r_n}{\|\varphi\|} = r_n.$$

Put $a := \varphi(y) + c_1 \in k$. We have just shown that $|a - c_{n+1}| \le r_n$, and $|c_{n+1} - c_n| \le r_n$ by (K2), so by the ultrametric inequality $|a - c_n| \le \max\big(|a - c_{n+1}|, |c_{n+1} - c_n|\big) \le r_n$ for every n. Thus $a \in \bigcap_n B(c_n, r_n)$, which contradicts (K2). □

This is one direction of van der Put's theorem [7, Théorème (3.2)]: k is spherically complete if and only if $(\ell^\infty/c_0)' \neq 0$. The next lemma, the description of the dual of ℓ^∞, is [7, Théorème (4.1)]; see also [8, Ch. 4]. We include a proof.

Lemma 13. *Let φ be a bounded linear functional on ℓ^∞. Then $\varphi(e_i) \to 0$ as $i \to \infty$, and $\varphi(x) = \sum_i x_i\varphi(e_i)$ for every $x \in \ell^\infty$. In particular $|\varphi(u)| \le \|u\| \sup_{i\ge N}|\varphi(e_i)|$ whenever $u = u|_{\ge N}$.*

Proof. Suppose that $\varphi(e_i) \not\to 0$. Then there are $\varepsilon_0 > 0$ and indices $i_0 < i_1 < \cdots$ with $|\varphi(e_{i_k})| \ge \varepsilon_0$. Let $T\colon \ell^\infty \to \ell^\infty$ be the operator with $T(x)(i_k) = x_k/\varphi(e_{i_k})$ and all other coordinates 0; it is linear and $\|T(x)\| \le \|x\|/\varepsilon_0$. So $\psi := \varphi \circ T$ is a bounded functional, and $\psi(e_k) = 1$ for all k because $T(e_k) = e_{i_k}/\varphi(e_{i_k})$. Let sh be the right shift, $\mathrm{sh}(x)(0) = 0$ and $\mathrm{sh}(x)(i+1) = x(i)$. Then $\psi \circ (1 - \mathrm{sh})$ is a bounded functional with

$$\big(\psi \circ (1 - \mathrm{sh})\big)(e_k) = \psi(e_k) - \psi(e_{k+1}) = 1 - 1 = 0 \qquad \text{for every } k,$$

so it vanishes by Proposition 12. But $(1 - \mathrm{sh})\mathbf{1} = e_0$, so $0 = \psi(e_0) = 1$, a contradiction.

Hence $x_i\varphi(e_i) \to 0$ for every $x \in \ell^\infty$. By the ultrametric inequality a difference of two partial sums of $\sigma(x) := \sum_i x_i\varphi(e_i)$ is bounded by its largest term, so the partial sums form a Cauchy sequence, and since k is complete the series converges, with $|\sigma(x)| \le \|x\| \sup_i|\varphi(e_i)| \le \|\varphi\|\|x\|$. So σ is a bounded functional with $\sigma(e_j) = \varphi(e_j)$ for all j, so $\varphi = \sigma$ by Proposition 12.

For the last assertion let $u = u|_{\ge N}$. Then $u_i = 0$ for $i < N$, so $\varphi(u) = \sum_{i\ge N} u_i\varphi(e_i)$. Every term of this series has absolute value at most $\|u\| \sup_{i\ge N}|\varphi(e_i)|$; by the ultrametric inequality so does every partial sum, and hence so does the limit $\varphi(u)$. □

Proposition 14 (multilinear Phillips property)**.** *Let $\lambda\colon (\ell^\infty)^d \to k$ be a continuous multilinear map and $\varepsilon > 0$. There is N such that $|\lambda(u_1,\dots,u_d)| \le \varepsilon$ whenever $\|u_j\| \le 1$ for all j and $u_j = u_j|_{\ge N}$ for some j.*

We call such an N a **threshold** for λ; every larger integer is then a threshold too.

Proof. Fix ε. By induction on d. For $d = 1$ choose N with $|\lambda(e_i)| \le \varepsilon$ for $i \ge N$ and apply Lemma 13.

Let $d \ge 2$ and assume the statement for forms in fewer than d variables. It suffices to show, for each j, that there is N_j such that $|\lambda(u_1,\dots,u_d)| \le \varepsilon$ whenever $u_1,\dots,u_d$ lie in the unit ball of ℓ^∞ and $u_j = u_j|_{\ge N_j}$; then $\max_j N_j$ is a threshold. Without loss of generality, assume $j = 1$. Suppose, for the sake of contradiction, that for every N there are $u_1,\dots,u_d$ in the unit ball with $u_1 = u_1|_{\ge N}$ and $|\lambda(u_1,\dots,u_d)| > \varepsilon$.

Step 1: *for every N there are $i \ge N$ and $w_2,\dots,w_d$ in the unit ball with $|\lambda(e_i, w_2,\dots,w_d)| > \varepsilon$.*

For such a tuple $u_1,\dots,u_d$, Lemma 13 applied to the bounded functional $\lambda(\cdot, u_2,\dots,u_d)$ gives $|\lambda(u_1,\dots,u_d)| \le \sup_{i\ge N}|\lambda(e_i,u_2,\dots,u_d)|$, so one of the terms on the right is $> \varepsilon$.

Define

$$\nu_{l,i}(v_1,\dots,v_{d-1}) := \lambda(v_1,\dots,v_{l-1},e_i,v_l,\dots,v_{d-1})$$

and $\mu_i := \nu_{1,i}$. These are forms in $d-1$ variables, so the inductive hypothesis applies to them.

Step 2: *construct indices $j_1 < j_2 < \cdots$, integers $0 = M_0 < M_1 < \cdots$ and vectors $z_2^{(s)},\dots,z_d^{(s)}$ in the unit ball, supported in $[M_{s-1}, M_s)$ such that*

$$|\mu_{j_s}(z_2^{(s)},\dots,z_d^{(s)})| = |\lambda(e_{j_s}, z_2^{(s)},\dots,z_d^{(s)})| > \varepsilon. \tag{2}$$

Suppose that stages $1,\dots,s-1$ have been carried out. Let K_s be a common threshold for the finitely many forms $\nu_{l,i}$ with $2 \le l \le d$ and $i < M_{s-1}$; for $s = 1$ we put $K_1 := 0$ and $j_0 := -1$. By Step 1 with $N = \max(K_s, j_{s-1}+1)$ there are $j_s \ge K_s$, with $j_s > j_{s-1}$, and $w_2,\dots,w_d$ in the unit ball with $|\mu_{j_s}(w_2,\dots,w_d)| > \varepsilon$. Let $M_s > M_{s-1}$ be a threshold for μ_{j_s}, and put

$$z_l^{(s)} := w_l|_{\ge M_{s-1}} - w_l|_{\ge M_s}, \qquad \text{so that} \qquad w_l = h_l + z_l^{(s)} + t_l,$$

where $t_l := w_l|_{\ge M_s}$ and $h_l = \sum_{i<M_{s-1}} c_{l,i}e_i$ with $|c_{l,i}| \le 1$. Expand $\mu_{j_s}(w_2,\dots,w_d)$ multilinearly into 3^{d-1} terms. A term containing some t_l has absolute value $\le \varepsilon$, because M_s is a threshold for μ_{j_s}. Consider a term containing no t_l but some h_l, say in slot l. By linearity in slot l, the term equals

$$\sum_{i<M_{s-1}} c_{l,i}\,\lambda(e_{j_s},\dots,\underbrace{e_i}_{\text{slot } l},\dots) = \sum_{i<M_{s-1}} c_{l,i}\,\nu_{l,i}(e_{j_s},\dots).$$

The first entry is still $e_{j_s} = e_{j_s}|_{\ge K_s}$, because $j_s \ge K_s$, and K_s is a threshold for every $\nu_{l,i}$ with $i < M_{s-1}$. So every summand has absolute value $\le \varepsilon$, and so does the term, by the ultrametric inequality. As $|\mu_{j_s}(w_2,\dots,w_d)| > \varepsilon$ while all these terms have absolute value $\le \varepsilon$, the ultrametric inequality shows that the remaining term $\mu_{j_s}(z_2^{(s)},\dots,z_d^{(s)})$ satisfies (2).

Step 3: because the supports $[M_{s-1}, M_s)$ of the $z_l^{(s)}$ are pairwise disjoint, and each $z_l^{(s)}$ lies in the unit ball, the sum $u_l := \sum_s z_l^{(s)}$ is well defined with $\|u_l\| \le 1$. Let $\chi := \lambda(\cdot, u_2,\dots,u_d)$, a bounded functional. Fix s and write $u_l = A_l + z_l^{(s)} + B_l$ with $A_l := \sum_{r<s} z_l^{(r)}$ and $B_l := \sum_{r>s} z_l^{(r)} = B_l|_{\ge M_s}$. Expand $\chi(e_{j_s}) = \mu_{j_s}(u_2,\dots,u_d)$ multilinearly. Exactly as before, with K_s and M_s as in stage s, the terms containing some B_l have absolute value $\le \varepsilon$, because M_s

is a threshold for μ_{j_s}; and the terms containing no B_l but some A_l have absolute value $\le \varepsilon$, because A_l is supported in $[0, M_{s-1})$ and K_s is a threshold for every $\nu_{l,i}$ with $i < M_{s-1}$. The remaining term is $\mu_{j_s}(z_2^{(s)}, \dots, z_d^{(s)})$. By (2) and the ultrametric inequality, $|\chi(e_{j_s})| > \varepsilon$ for every s. Since $j_s \to \infty$, this contradicts Lemma 13. □

This is where (K2) is used. The statement fails on c_0, where $x \mapsto \sum_i x_i$ is a bounded functional taking the value 1 at every e_i, and it fails on ℓ^∞ over a spherically complete field, where the Hahn–Banach theorem [3] gives a nonzero bounded functional vanishing on c_0.

6. Step 4: gluing

Recall from Step 1 that Φ is 4-linear with values in F and that

$$\|\Phi(a,b;x,y)\| \le C\|a\|\,\|b\|\,\|x\|\,\|y\| \qquad \text{with } C \ge 1.$$

We now combine Corollary 11 (Step 2) with Proposition 14 (Step 3).

Choose inductively finite sets $I_1 < I_2 < \cdots$ in $\mathbb{N}$ (every element of I_t is smaller than every element of I_{t+1}) with $|I_t| = 2t$ and vectors $a_t, b_t \in \{0,1\}^{I_t}$ as follows. Given the earlier stages let $a' = \sum_{s<t} a_s$, $b' = \sum_{s<t} b_s$, and apply Corollary 11 to get a_t, b_t with $|\det G_t| \ge 1$, where $G_t = \Phi(a' + a_t, b' + b_t; a' + a_t, b' + b_t)|_{I_t \times I_t}$. Put $\varepsilon_t := \frac{1}{2} C^{-2t}$ and use Proposition 14 (with $d = 4$) for the finitely many entry forms $\Phi(\cdot)_{pq}$, $p, q \in I_t$, to get N_t such that these forms are at most ε_t on unit arguments one of which is supported in $[N_t, \infty)$. All later sets I_s, $s > t$, are chosen inside $[N_t, \infty)$.

Let $a, b \in \ell^\infty$ be defined by $a|_{I_t} = a_t$, $b|_{I_t} = b_t$ and zero elsewhere. Then $a = \mathbf{1}_A$ and $b = \mathbf{1}_B$ are indicator functions of two subsets of $\mathbb{N}$, so m_a, m_b are coordinate projections. Let $\mathcal{O} := \Phi(a,b;a,b) \in F$. Fix t and split $a = a^{\le t} + a^{>t}$ and $b = b^{\le t} + b^{>t}$, where $a^{\le t} = a' + a_t$ and $b^{\le t} = b' + b_t$ are the vectors of stage t, and $a^{>t}, b^{>t}$ are supported in $[N_t, \infty)$. Expand $\mathcal{O}$ multilinearly into 16 terms. The term containing neither $a^{>t}$ nor $b^{>t}$ restricts to G_t on $I_t \times I_t$; each of the other 15 terms has entries of absolute value $\le \varepsilon_t$ there, by the choice of N_t. So $\|\mathcal{O}|_{I_t \times I_t} - G_t\| \le \varepsilon_t$ by the ultrametric inequality. All entries are bounded by C. Expanding the determinant multilinearly in columns, every term containing a column of the perturbation is at most $\varepsilon_t C^{2t-1} < 1$, so the ultrametric inequality gives

$$|\det \mathcal{O}|_{I_t \times I_t}| = |\det G_t| \ge 1 \qquad \text{for every } t.$$

On the other hand, every element of F is a uniform limit of matrices of finite rank, because each $W(x,y)$ has rank ≤ 2. So $\mathcal{O} \in F$ is within $\eta := (2C)^{-1}$ of a matrix R of some finite rank r. Expanding $\det(R + (\mathcal{O} - R))$ on $I_t \times I_t$ in the same way, terms with more than r columns from R vanish, so $|\det \mathcal{O}|_{I_t \times I_t}| \le C^r \eta^{2t-r} < 1$ once $t > r$. This contradiction proves Theorem 2. □

Appendix A. An explicit field

The purpose of this appendix is to construct a field satisfying (K1) and (K2). Towards that end, let $k_0 = \mathbb{F}_2$ (or any field of characteristic 2). For $j \ge 0$ let $k_0((t^{1/2^j}))$ be the field of *formal* Laurent series $\sum_{n \ge N} a_n t^{n/2^j}$ ($N \in \mathbb{Z}$, $a_n \in k_0$) in the single variable $t^{1/2^j}$. These fields form an increasing chain, and we let

$$\mathcal{P} := \bigcup_{j \ge 0} k_0((t^{1/2^j})), \qquad |f| := 2^{-\operatorname{ord} f},$$

be the field of **2-power Puiseux series**. Here $\operatorname{ord} f$ is the minimum q for which $\operatorname{coeff}_{t^q} f \ne 0$. We set $|0| = 0$ by definition. Let k be the completion of $\mathcal{P}$ for this (non-Archimedean) absolute value. This is characteristic 2 since $k_0 \hookrightarrow \mathcal{P} \hookrightarrow k$.

Proposition 15. *k is not spherically complete.*

Proof. We exhibit a sequence of balls as in (K2). Put

$$\gamma_m := 1 - 2^{-(m+1)}, \qquad c_n := \sum_{m=0}^{n} t^{\gamma_m} \in \mathcal{P}, \qquad r_n := 2^{-\gamma_{n+1}}.$$

The balls have the form required in (K2). Since $c_{n+1} - c_n = t^{\gamma_{n+1}}$ we have $|c_{n+1} - c_n| = 2^{-\gamma_{n+1}} = r_n$. Since the γ_m are strictly increasing to 1, the r_n are strictly decreasing, and $r_n > 2^{-1} = \frac{1}{2}$.

The intersection is empty. Suppose that $a \in k$ satisfies $|a - c_n| \le r_n$ for all n. As $\mathcal{P}$ is dense in k there is $y \in \mathcal{P}$ with $|a - y| < \frac{1}{2}$, and then for every n

$$|y - c_n| \le \max\big(|y - a|, |a - c_n|\big) \le \max\big(\tfrac{1}{2}, r_n\big) = r_n. \tag{3}$$

Now $y \in k_0((t^{1/2^j}))$ for some j. The exponent $\gamma_j = (2^{j+1} - 1)/2^{j+1}$ has an odd numerator, therefore γ_j is not a multiple of $1/2^j$ and so $\operatorname{coeff}_{t^{\gamma_j}} y = 0$. For $n \ge j$ we have $2^{-\gamma_j} > 2^{-\gamma_{n+1}} = r_n$. Furthermore, the term t^{γ_j} occurs in c_n, so

$$\operatorname{coeff}_{t^{\gamma_j}}(y - c_n) = 0 - 1 \neq 0, \qquad \text{hence} \qquad \operatorname{ord}(y - c_n) \le \gamma_j \quad \text{and} \quad |y - c_n| \ge 2^{-\gamma_j} > r_n,$$

which contradicts (3). □

Appendix B. Formalization

The Lean 4/Mathlib development is at

https://github.com/Deicyde/alternating-analytic-counterexample

The library `AltCounter/` contains no `sorry`, and the only axioms it uses are the three standard ones (`propext`, `Quot.sound` and `Classical.choice`). Its headline theorem assumes that Mathlib's

`ContinuousAlternatingMap.compContinuousLinearMapCLM`

is C^ω for all nontrivially normed fields, normed spaces and finite index types, and derives `False`; a second theorem does the same assuming this only for complete non-Archimedean fields of characteristic 2, Banach spaces, $E' = E$ and $|\iota| = 2$. Both are restated in a Mathlib-only file `Challenge.lean` (with `sorry` in place of the proofs, as that format requires) and checked against `Solution.lean` by the `leanprover/comparator` tool.

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